The Generalized Rupert Problem and its Energy Landscapes

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–This will be written last – A geometry problem that dates back to the seventeenth-century exploits of a European prince is shown to have an intriguing self-similar structure.

There are difficulties in jamming objects into confined spaces. Objects of similar size and shape can induce frustration, and corners in spaces can be inaccessible (while at the same time providing novel constraints). Optimal packings of nonoverlapping objects have been puzzled over for thousands of years, and centuries have passed without substantial progress on certain problems. For instance, the densest arrangement of equal-sized regular tetrahedra in 3-space is still unknown [Torquata/Conway, Glotzer, Cohn]. In recent years, theoretical ideas and computational techniques from physics, such as simulated annealing, parallel tempering, molecular dynamics[2, 3], and models of quasicrystalline order[**?**], have re-energized some of these age-old problems. The subjects of covering, filling and jamming have emerged from attempts to reconcile geometrical constraints and energy minimization. In this Letter, we take a physics approach to a problem that stems from the 17th-century exploits of the European prince, Rupert of the Rhine, namely that of finding the largest square that fits entirely inside a unit cube [**?**].

Rupert's problem: A short history. In the mid-17th century, Prince Rupert (1619-1682), also called Ruprecht Pfalzgraf bei Rhein, and someone known for both military and scientific inventions, posed the following geometrical puzzle: Can one form a hole in a cube such that a second cube of the same size may pass through the hole? This was the original Rupert problem, and its answer is "yes". John Wallis [wallis], placing a square hole on the plane normal to the cube's main diagonal, found that the square hole could have sidelength as large as $\sqrt{6}-\sqrt{2}=1.03527....$ Wallis thus solved the original problem; he also thought his solution was maximal, but he was wrong. The optimal square hole lies on a plane with lesser symmetry, as discovered by Pieter Nieuwland, in the late 18th-century. He calculated a maximal sidelength of $\sqrt{9/8}=1.06066...$, which is equivalent to the side of the largest square in a unit cube (Fig. 1a). Centuries passed, then Martin Gardner, in a 1965 column devoted to hypercubes [gardner-sciam], asked for the sidelength of the largest 3-cube inscribable in a unit 4-cube. A number of the solutions elicited by Gardner's article were correct: the sidelength $f(3,4)$ of the 3-cube in the 4-cube is approximately 1.00743... (XXX first discovered this value, but it has been called "DeVicci's constant" [14] primarily because DeVicci was the first to provide a proof [**?**].)

Generalized Rupert problem. It is but a short step to ask for the sidelength, $f(n-1,n)$, of the largest $(n-1)$-cube inscribable in an n-cube. The resulting numerics are so intriguing that we display them here for $n<16$, even before discussing our methods for computing them:

$$
\begin{aligned}
f(1,2) &= 1.41421356237309504880168 87\\
f(2,3) &= 1.06066017177982128660126 65\\
f(3,4) &= 1.00743475688427937609825 36\\
f(4,5) &= 1.00083944685934978860192 89\\
f(5,6) &= 1.00009347357651963189100 75\\
f(6,7) &= 1.00001038883740650822613 04\\
f(7,8) &= 1.00000115435563441097530 14\\
f(8,9) &= 1.00000012826229403987533 03\\
f(9,10) &= 1.00000001425137360172276 78\\
f(10,11) &= 1.00000000158348605845609 20\\
f(11,12) &= 1.00000000017592089676208 96\\
f(12,13) &= 1.00000000001954921076971 54\\
f(13,14) &= 1.00000000000217213453021 20\\
f(14,15) &= 1.00000000000024134828113 79
\end{aligned}
$$

Although the Babylonians had the solution $f(1,2)=\sqrt{2}$[**?**], it is still worth discussing because: 1) features of this geometry have similarities to the general problem, 2) a bound on $f(n-1,n)$ can lead to a bound on $f(n,n+1)$, and 3) a solution to $f(n-1,n)$ can be perturbed to build a solution to $f(n,n+1)$.

By examining this family of numerical lower bounds, one can see the asymptotic law $f(n-1,n)\sim 1+\frac{1}{2}9^{2-n}$ [need to fix this, if it is not quite correct]. We'll see that asymptotics of this type, and of other forms, arises in other families as well.

It is but another short step to consider the general problem of $f(m,n)$: the sidelength of the largest m-cube that fits inside a unit n-cube.

Matrix representations. There are three separate systems of matrices to keep in mind in the general problem: vertices (V), edge vectors (E) and rotations (R). Henceforth, when discussing a matrix, we'll indicate if it is a V, E or R matrix.

For concreteness, we will look at V and E in the case of the cube in a tesseract, $(m,n)=(3,4)$.

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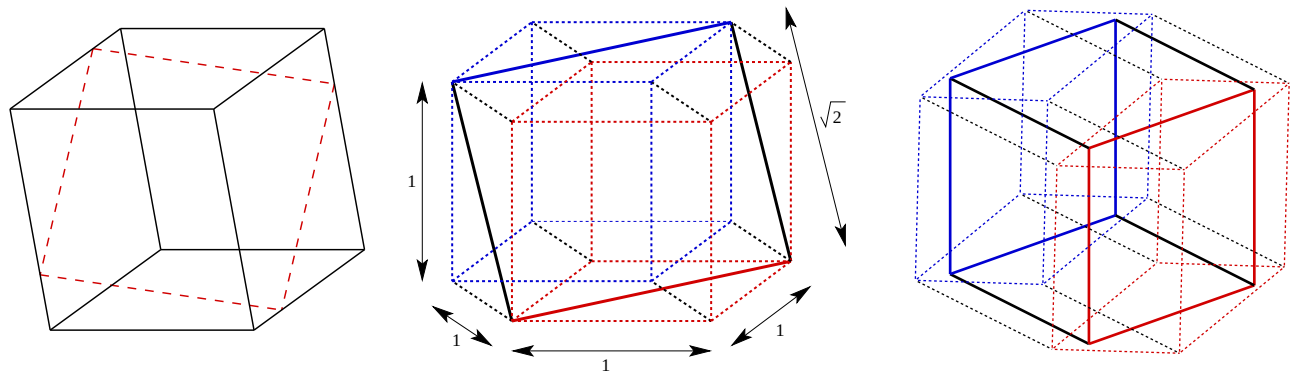

FIG. 1. (a) Nieuwland's optimal solution to Rupert's original problem where the square's vertices divide the cube's edges in the length ratio of 1:3. (b) The maximal square within a tesseract. (c) The maximal cube within a tesseract.

Cracking the DeVicci Cube. After some empirical investigation, one can guess that vertex matrix, V, for $(m, n) = (3, 4)$ is of the form:

$$V = \begin{bmatrix} v_{0,0,0} \\ v_{1,0,0} \\ v_{0,1,0} \\ v_{1,1,0} \\ v_{0,0,1} \\ v_{1,0,1} \\ v_{0,1,1} \\ v_{1,1,1} \end{bmatrix} = \begin{bmatrix} x & 1/2 & y & 1/2 \\ 1/2 & 1/2 & z & -w \\ z & -1/2 & 1/2 & w \\ y & -1/2 & x & -1/2 \\ -y & 1/2 & -x & 1/2 \\ -z & 1/2 & -1/2 & -w \\ -1/2 & -1/2 & -z & w \\ -x & -1/2 & -y & -1/2 \end{bmatrix}$$

This makes the edge vector matrix, E:

$$E = \begin{bmatrix} e_{1,0,0} \\ e_{0,1,0} \\ e_{0,0,1} \end{bmatrix} = \begin{bmatrix} -x+1/2 & 0 & -y+z & -w-1/2 \\ -x+z & -1 & -y+1/2 & w-1/2 \\ -x-y & 0 & -y-x & 0 \end{bmatrix}$$
$$= \begin{bmatrix} a & 0 & b & c \\ d & -1 & a+b-d & -c-1 \\ 2a+b-d-1 & 0 & 2a+b-d-1 & 0 \end{bmatrix}$$

where $a = -x + 1/2$, $b = -y + z$, $c = -w + 1/2$, and $d = -x + z$. Using the facts that the edge vectors have equal length, l, and are mutually orthogonal, we obtain the following system of equations:

$$\begin{aligned} 2a^2 + c^2 &= l^2 \\ 2ad - c(c+1) &= 0 \\ 2d^2 + (c+1)^2 + 1 &= l^2 \\ 2(a-d-1)^2 &= l^2 \end{aligned}$$

Letting $t = l^2$, these equations can be reduced to:

$$\begin{aligned} p(t) = 16t^8 - 224t^7 + 984t^6 - 1528t^5 + 625t^4 \\ + 352t^3 + 32t^2 - 512t + 256 = 0 \end{aligned}$$

and the solution of the original problem is the smallest, positive, real root. This is:

$$\begin{aligned} t &= \frac{1}{12}\left(21 + \sqrt{3c} - \sqrt{1449 + 6480\sqrt{\frac{3}{c}} - 3c}\right) \\ &= 1.0149247893... \end{aligned}$$

$$\begin{aligned} c = 161 &+ (184841 - 4320\sqrt{1290})^{1/3} \\ &+ (184841 + 4320\sqrt{1290})^{1/3} \end{aligned}$$

and this means $f(3, 4) \geq l = 1.0074347568...$

This is a lower bound on $f(3, 4)$. DeVicci was the first to provide a proof [**?**] that this bound was the solution. Her proof is very long and contains many, many cases. This approach to finding bounds for $f(m, n)$ and proving they are maximal is problematic. We adopted a different approach to exploring $f(m, n)$, both numerically and mathematically.

Rotation Matrices and SO(n). Another way to view the problem is given an axis aligned n-cube and an axis aligned m-cube in n-dimensions with all vertex coordinates above m set equal to zero, what is the rotation matrix and scaling that takes the canonical m-cube an optimal m-cube for this problem? The rotation matrix, R, is simply an element of SO(n) and the scaling is a real number, l. Once you know R, l can easily be computed, thus only R needs to be found.

R is an n x n orthonormal matrix with entries $r_{i,j}$. The generalized Rupert problem then becomes:

$$f(m, n) = \left(\min_{R \in SO(n)} \left(\max_{i=1}^{n} \sum_{j=1}^{m} |r_{i,j}|\right)\right)^{-1}$$

Fig. 2 shows how the partial row sums behave and enter into the max part of the min-max problem for the (1,2) and (2,3) cases. Discuss the characteristics of the candidates, the max of these candidates, and the global optima.

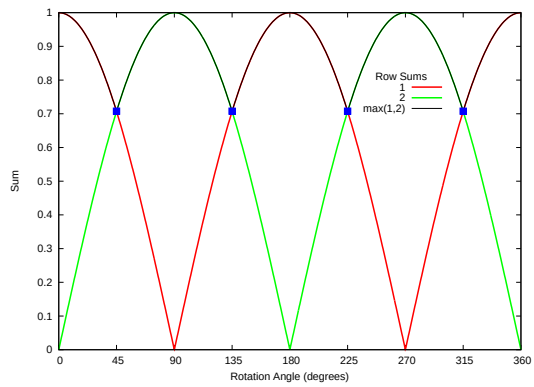

FIG. 2. (a) Row-sums as a function of angle space for (1,2) problem. (b) Row-sums as a function of angle space for (2,3) problem.

Transition into the numerical techniques we'd used to explore the overall problem. Metropolis/simulated annealing.

Simulated annealing. Simulated annealing is a Monte Carlo approach for minimizing such multivariate functions. The term simulated annealing derives from the roughly analogous physical process of heating and then slowly cooling a substance to obtain a strong crystalline structure. In simulation, a minima of the cost function corresponds to this ground state of the substance. The simulated annealing process lowers the temperature by slow stages until the system "freezes" and no further changes occur. At each temperature the simulation must proceed long enough for the system to reach a steady state or equilibrium. This is known as thermalization. The time required for thermalization is the decorrelation time; correlated microstates are eliminated. The sequence of temperatures and the number of iterations applied to thermalize the system at each temperature comprise an annealing schedule. To apply simulated annealing, the system is initialized with a particular configuration. A new configuration is constructed by imposing a random displacement. If the energy of this new state is lower than that of the previous one, the change is accepted unconditionally and the system is updated. If the energy is greater, the new configuration is accepted probabilistically. This is the Metropolis step, the fundamental procedure of simulated annealing. This procedure allows the system to move consistently towards lower energy states, yet still jump out of local minima due to the probablistic acceptance of some upward moves. If the temperature is decreased logarithmically, simulated annealing guarantees an optimal solution.

The original Metropolis scheme was that an initial state of a thermodynamic system was chosen at energy E and temperature T, holding T constant the initial configuration is perturbed and the change in energy dE is computed. If the change in energy is negative the new configuration is accepted. If the change in energy is positive it is accepted with a probability given by the Boltzmann factor $exp(\Delta E/T)$. This processes is then repeated sufficient times to give good sampling statistics for the current temperature, and then the temperature is decremented and the entire process repeated until a frozen state is achieved at $T = 0$.

By analogy the generalization of this Monte Carlo approach to matrix (geometric? combinatoric?) problems is straight forward The current state of the thermodynamic system is analogous to the current solution to the combinatorial problem, the energy equation for the thermodynamic system is analogous to at the objective function, and ground state is analogous to the global minimum. The major difficulty (art) in implementation of the algorithm is that there is no obvious analogy for the temperature T with respect to a free parameter in the combinatorial problem. Furthermore, avoidance of entrainment in local minima (quenching) is dependent on the "annealing schedule", the choice of initial temperature, how many iterations are performed at each temperature, and how much the temperature is decremented at each step as cooling proceeds.

Bootstrapping. Going from an optimal (m,n) solution to a higher dimensional solution by using local minimum finding techniques.

Next, I decided to look at the 3 in 7-cube (which also had 4 positional unknowns and the side length):

$$\begin{aligned} v &= (-1, -x, -y, -1, -1, -1, -1) \\ v1 &= (-1, -y, -x, -1, 1, -1, 1) \\ v2 &= (-z, -w, -1, 1, -1, 1, -1) \\ v3 &= (z, -1, w, -1, -1, -1, -1) \end{aligned}$$

This looks structurally very similar and gave a similar looking set of equations but the solution (via Maple) is a bit more complicated, z is the minimum real solution to:

$$z^8 - 12z^6 + 4z^5 - 6z^4 + 120z^3 - 208z^2 + 300z - 151 = 0$$
$$z \approx 0.7227902063$$

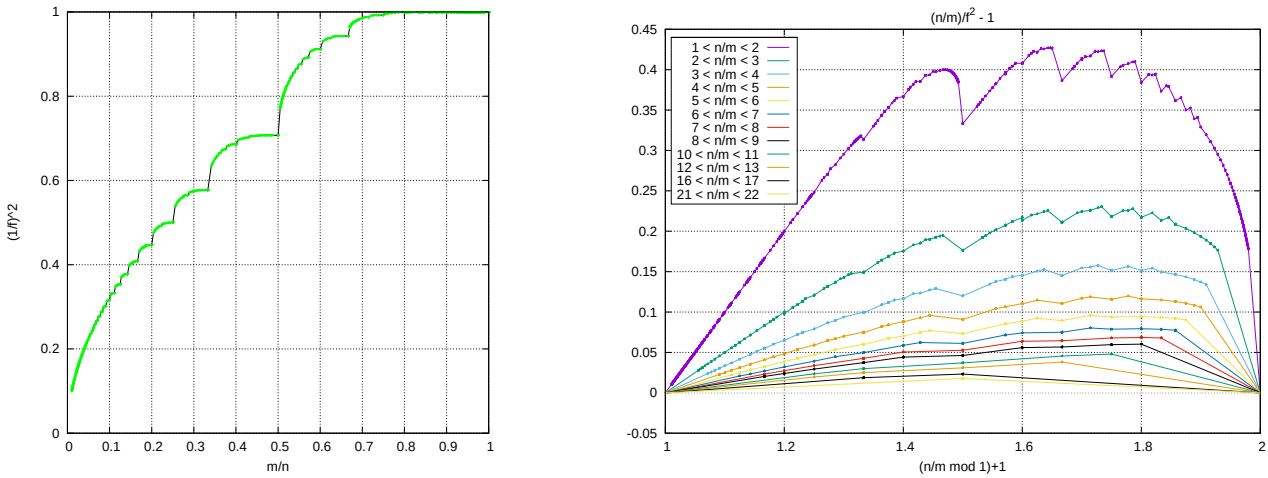

FIG. 3. (a) Inverse square of the edge length as a function of m/n. (b) $1/f^2 - n/m$ versus $(n/m \bmod 1) + 1$.

$$
\begin{aligned}
x &= \frac{1}{456328}(625456 - 484389z + 96148z^2...) \\
y &= \frac{1}{456328}(589343 - 413126z + 286081z^2...) \\
w &= \frac{1}{456328}(492441 - 71263z - 189933z^2...) \\
l &= \sqrt{2(x+y)^2 + 8}
\end{aligned}
$$

The expressions for x, y, and w continue to z^7 terms with the coefficients (mostly) decreasing in magnitude. Since $0 < z < 1$, this sort of looks like some type of truncated (convergent?) power series. The numerical results for these expressions give results consistent with the numerics.

Then I decided to look at some of the easy examples (one positional unknown and the side length). First, I looked at the 3 in 10-cube:

$$
\begin{aligned}
v &= (-1, -x, -x, -1, -1, -1, -1, -1, -1, -1) \\
v1 &= (-1, -1, -x, -x, 1, -1, 1, -1, 1, -1) \\
v2 &= (-x, -x, -1, 1, -1, 1, -1, 1, -1, 1) \\
v3 &= (x, 1, 1, x, -1, -1, -1, -1, -1, -1)
\end{aligned}
$$

In this case, all the orthogonal equations are trivially true and two the length equations are identical. This is what's left:

$$
\begin{aligned}
l^2 &= 2(x-1)^2 + 12 \\
l^2 &= 4(x+1)^2
\end{aligned}
$$

$$
\begin{aligned}
x &= -3 + \sqrt{14} \\
l &= 2(x+1) = l(0)(seebelow)
\end{aligned}
$$

After this a 3 in (3n+10)-cube results have the same form by appending n copies of (-1,-1,-1) to v, (-1,-1,1) to $v1$, (-1,1,-1) to $v2$, and (1,-1,-1) to $v3$. The length is given by: $l(n)^2 = l(0)^2 + 4n$.

To try to complete the 3 in q-cube case, I decided to look at the 3 in (3n+5)-cube. For $n = 0$ (5-cube), this gave rise to:

$$
\begin{aligned}
l^2 &= 2(x+1)^2 \\
l^2 &= 4(x-1)^2 + 4
\end{aligned}
$$

$$
\begin{aligned}
x &= 3 - \sqrt{6} \\
l &= \sqrt{2}(x+1) = l(0)
\end{aligned}
$$

and, in general, $l(n)^2 = l(0)^2 + 4n$.

Finally, this made me notice that there seemed to be pattern in the m in (2m-1)-cube cases. Namely, there always seemed to be only one coordinate unknown, the coordinates/vectors had a very simple form, and they always resulted in two equations in the two unknowns (with orthogonality guaranteed):

$$
\begin{aligned}
s(1) &= (x+1, x+1, 0, 0, 0, ..., 0, 0, 0) \\
s(2) &= (0, 0, x+1, x+1, 0, ..., 0, 0, 0) \\
&... \\
s(m) &= (0, 0, 0, 0, 0, ..., x+1, x+1, 0) \\
s(m+1) &= (-x+1, x-1, -x+1, x-1, -x+1, \\
&\qquad ..., -x+1, x-1, 2)
\end{aligned}
$$

$$
\begin{aligned}
l^2 &= 2(x+1)^2 \\
l^2 &= (2n-2)(x-1)^2 + 4
\end{aligned}
$$

$$
\begin{aligned}
x &= \frac{\sqrt{n}}{\sqrt{n} + \sqrt{2}} \\
l &= \sqrt{2}(x+1) = l(0)
\end{aligned}
$$

And, as in previous cases, this extends to the m in (mn + (2m-1))-cube case: $l(n)^2 = l(0)^2 + 2n$ for all n and m.

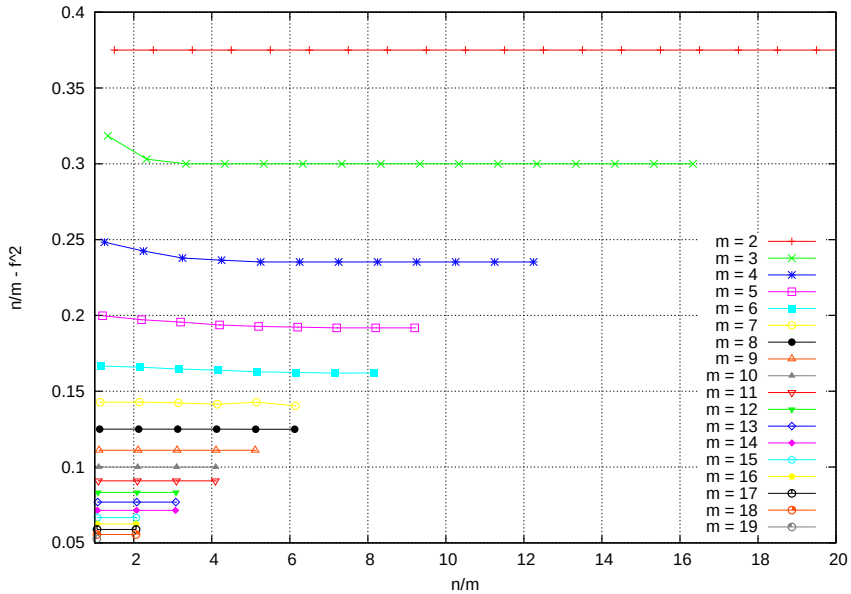

FIG. 4. Energies of families of solutions as a function of n/m.

Fractal frustration. What is the quantity that measures the "frustration" of a cube? Which cases of (m, n) are the most frustrated? How can frustration be read off the graph of $1/f^2$ versus m/n?

Fig. 3a is $1/l^2$ versus m/n, and Fig. 3b is $n/l^2 - 1$ versus n/m graph with all the lobes shifted into an interval of length 1, and some appropriate closeups. Discuss our overall impression that there these graphs/point-sets are self-similar, suggest methods for superimposing them, and discuss the possible significance.

3-cubes in n-cubes

The first truly new result that came out of the computer results was a pattern in the direction cosines for the largest 3cube in a 5cube. The pattern was blah blah blah

This pattern translates in another pattern for the vertices of the 3-cube:

$$\begin{array}{ccccc}
-\alpha & \frac{1}{2} & -\alpha & -\frac{1}{2} & -\frac{1}{2} \\
-\frac{1}{2} & \alpha & -\frac{1}{2} & -\alpha & \frac{1}{2} \\
-\alpha & \frac{1}{2} & \frac{1}{2} & \alpha & -\frac{1}{2} \\
-\frac{1}{2} & \alpha & \alpha & \frac{1}{2} & \frac{1}{2} \\
\alpha & -\frac{1}{2} & \alpha & \frac{1}{2} & \frac{1}{2} \\
\frac{1}{2} & -\alpha & \frac{1}{2} & \alpha & -\frac{1}{2} \\
\alpha & -\frac{1}{2} & -\frac{1}{2} & -\alpha & \frac{1}{2} \\
\frac{1}{2} & -\alpha & -\alpha & -\frac{1}{2} & -\frac{1}{2}
\end{array}$$

where $\alpha = 0.275255128...$ This leads to an edge length $= \sqrt{11 - 4\sqrt{6}} = 1.09637631717731....$

Here is how successive generations of computer experiments approached this value:

1.0963763169444
1.0963763170196
1.0963763171015
1.0963763171713
1.0963763171768

Extracting families of (m,n) from the previous graphs. Discuss their organization, analogies to energy level curves, and their asymptotics. Introduce the idea of "family extensions" - taking known solutions and using them to construct lower bound candidates for unknown solutions. Discuss using them to produce non-constructive lower bounds for unknown solutions (e.g. using f(m,n) inequalities). Discuss cases where the optimal solution candidates obtained by 4(b) seem to correspond to the "family extensions" after some point in the family (e.g., the (3,5) family).

In the bound on the $(m-1, m)$ case, the number $2 \cdot 3^{2m-4}$ appears somewhat naturally. Is this indicative of an asymptotic exponential form for the "correction" to 1 as m goes to infinity. An early, crude fit to the data for small m gave the following:

$$f(m-1, m)^2 \approx 1 + \frac{67.3}{8.2^m}$$

Notice that this expression is very nearly $1 + 8.2^{2-m}$ which is eerily reminiscent of the expression from the bound: $1 + \frac{1}{2}\, 9^{2-m}$.

Analytic results for $(m-1, m)$ and $(m-2, m)$ (lower bounds) We have been able to construct systems of equations for the $(m-1, m)$ family and the $(m-2, m)$ family. We have solved these systems iteratively using Mathematica to arbitrary precision.

Other families This line of thinking suggests other "families" of cases. For example, a Fibonacci family: f(1,2), f(2,3), f(3,5), f(5,8), etc. These values should approach a unique answer what is it? Every irrational can be approached in a similar way....what are the asymp-

totics of the approach to these irrationals? does it depend on the type of irrational?

$$f(m,n) = ... \tag{1}$$

For example, Eq.(1) refers to the side length of the m-cube in the problem.

Future questions – taken from our last meeting

Can we use the Farey tree to organize each of the m/n problems, or to organize the modular classes?

Does the modular table capture the true structure of the problem? Are there hidden families in the table that need to be exposed? Do they shed light on the full problem?

How are the cases $m > n/2$ related to the cases $m < n/2$? What exactly is the transformation? What else can we say about the fractal or self affine properties of the $1/f^2$ plot?

Are there analogs of energies or eigenvalues or ... in this problem? Are there other quantities besides $1/f$ associated with the global solution?

Using the systems of equations for $(m-1, m)$ and $(m-2, m)$ can the asymtotics be shown? Can any of this be proven to be a maximizer instead of simply a lower bound?

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[14] book, papers
[15] thesis?
[16] a number of papers
[17] PNAS
[18] frustration, jamming references
[19] jamming phase diagram
[20] dimension counting, field theory T.Lubensky
[21] j. wetzel papers